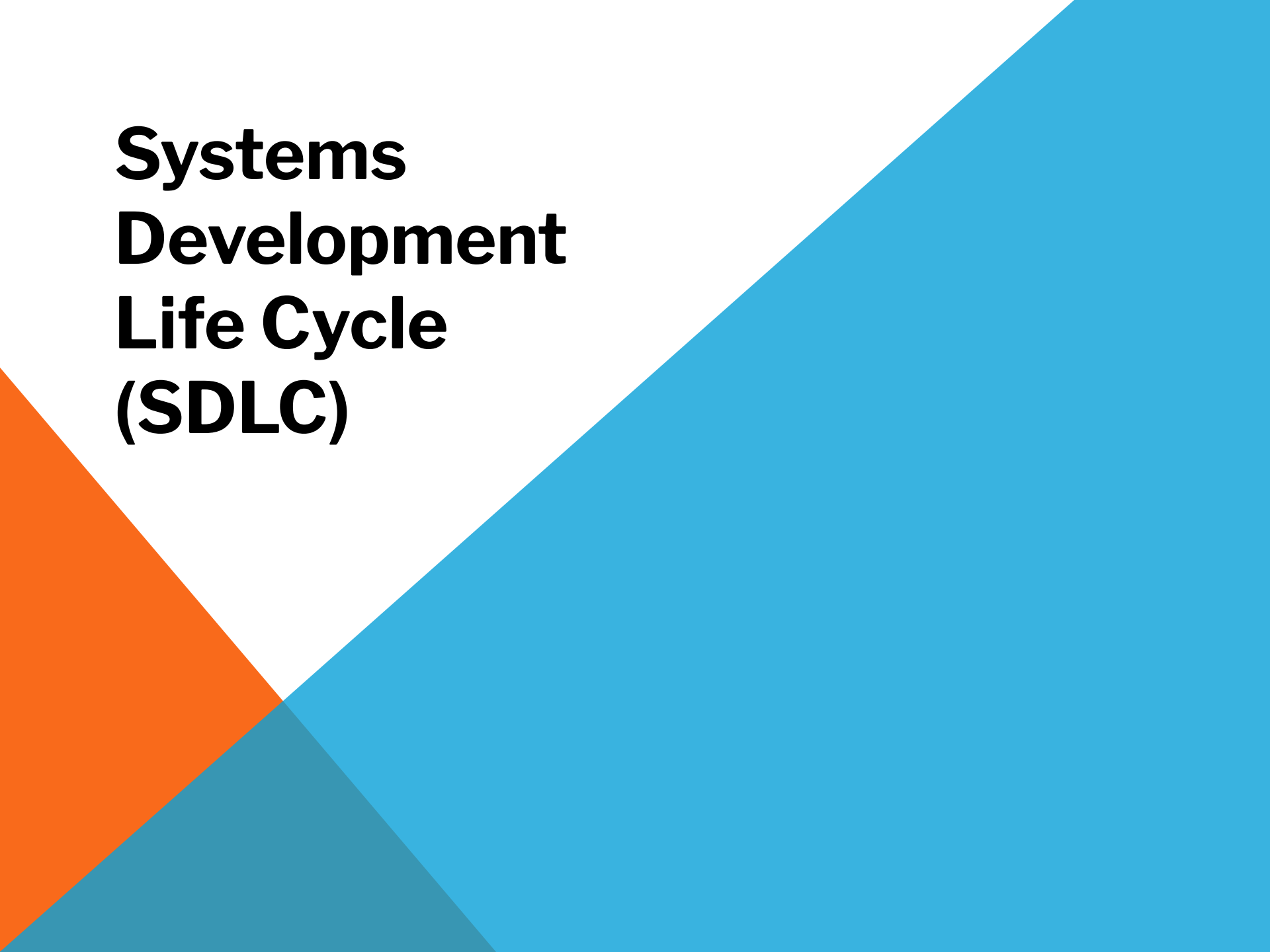




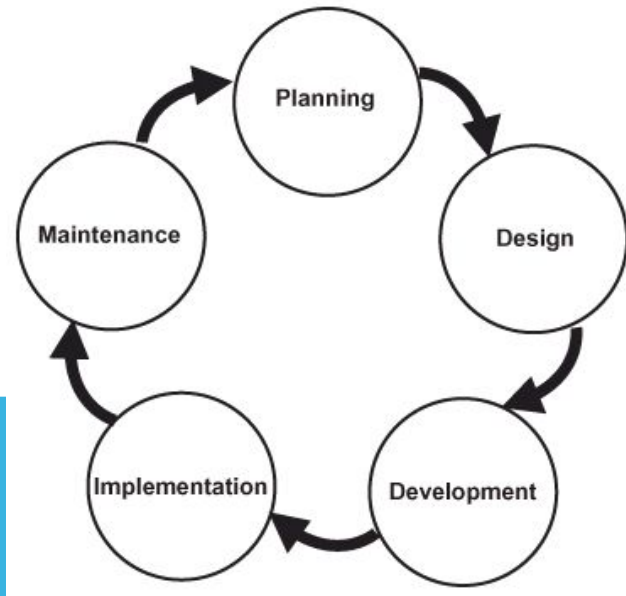
Systems Development Life Cycle (SDLC)

WHAT IS SDLC?

- Is a process of creating or modifying information systems, the models and methodologies.
 - SDLC concept supports many kinds of software development methodologies
 - the methodologies create the framework for planning and controlling the creation of an information system.
 - Clients/User need to be wary that the design of good applications software has to go through a process of procedural stages.
- 

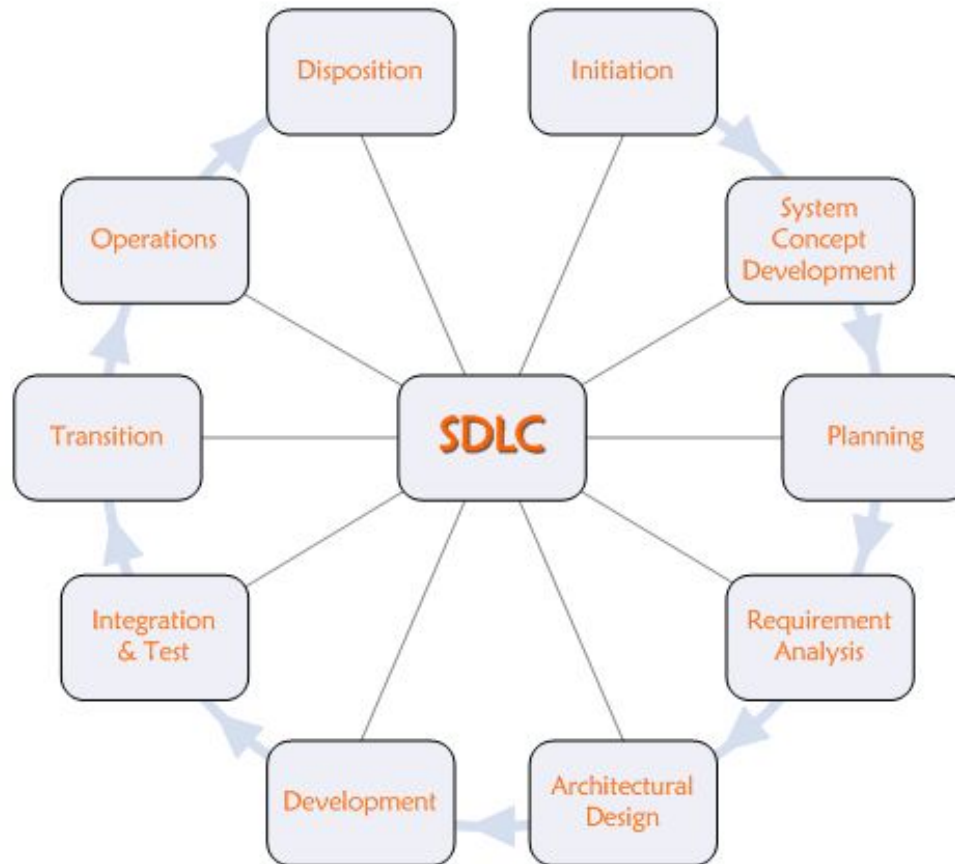
SDLC

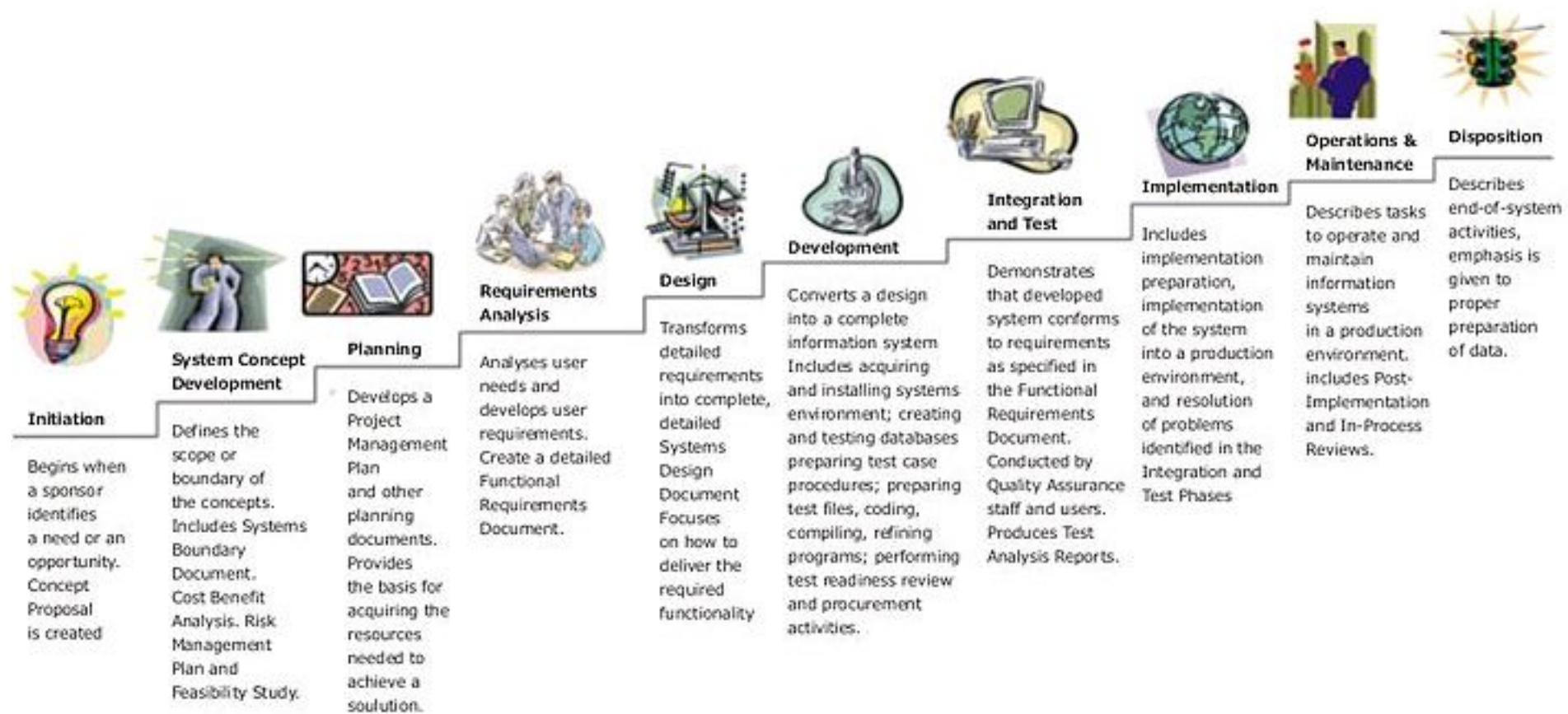
Information gathered in the planning phase is used to design the content of table structures, the data fields, the data types in each field and more importantly in a relational database, the one-to-many relationships that need to link the many tables in the organized database structure.



SDLC FRAMEWORK DIAGRAM

Systems Development Life Cycle





SDLC

MODELS/METHODOLOGIES

- Waterfall
- Prototype
- Agile



WATERFALL MODEL

Advantages

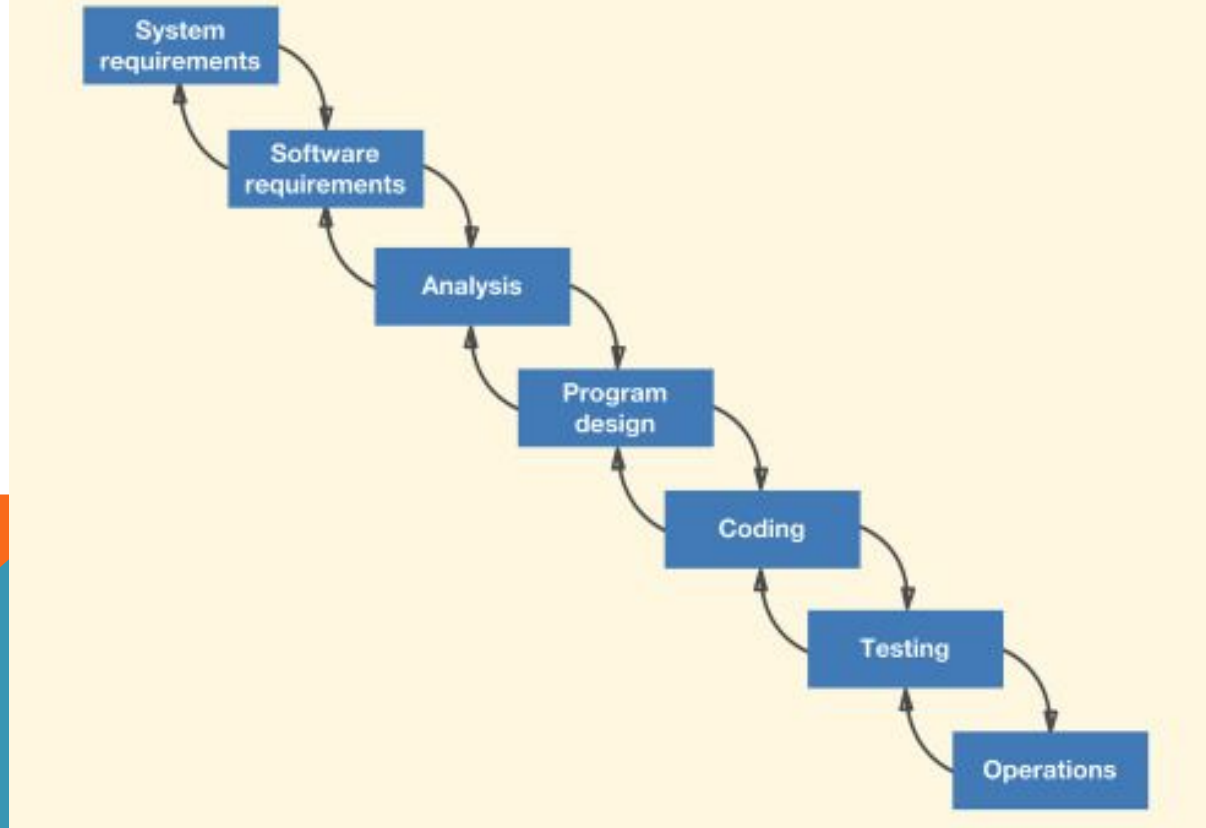
- Save time and cost
- Structured approach
- Easy to understand and explain
- Easy to identify milestones

Disadvantages

- Has bugs in the early stage of design
- Less details in design and documentation
- More challenging for the project team to follow-up when there is a change of team member
- Lack of flexibility

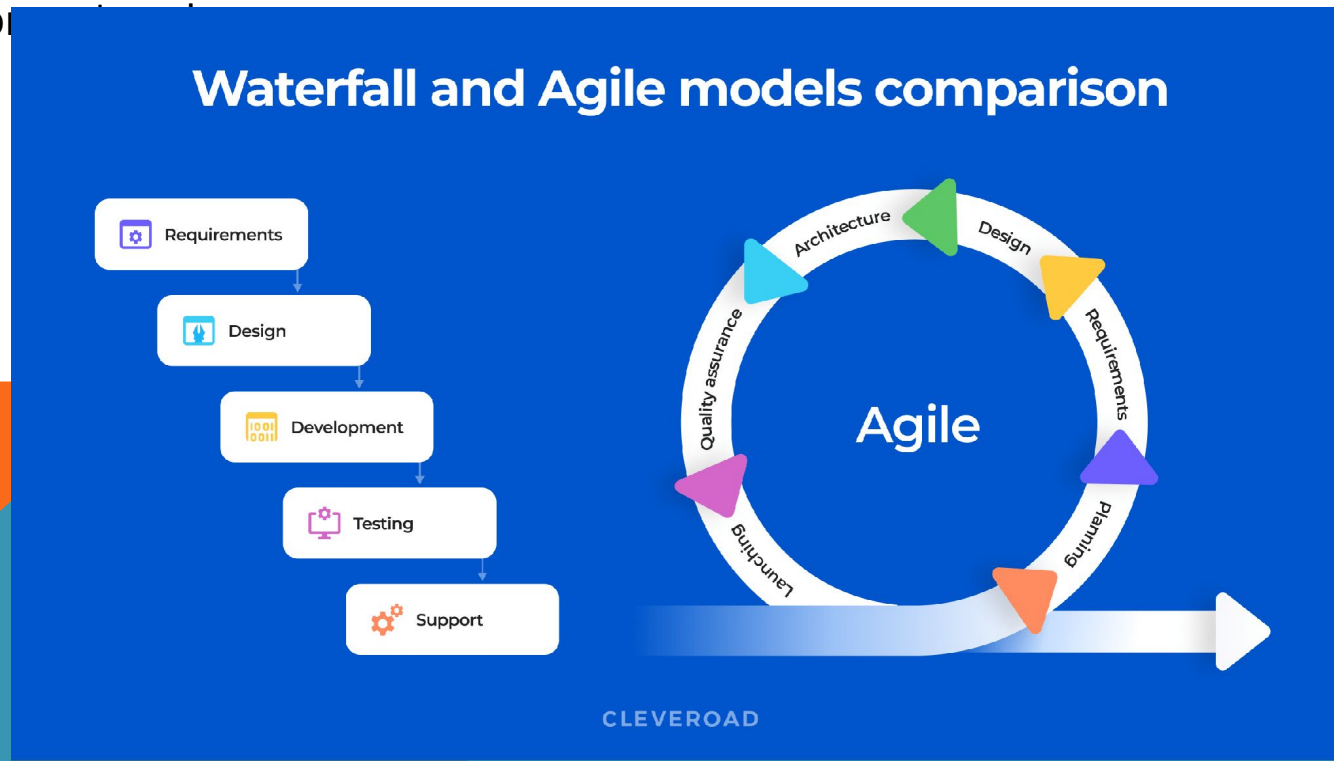
WHAT IS WATERFALL MODEL?

- Often used in software processes and the industry of construction and manufacturing
- Has a progress flows in the phases of CONCEPTION, INITIATION, ANALYSIS, DESIGN , CONSTRUCTION, TESTING , PRODUCTION, and MAINTENANCE.
- And the progress flows firmly from the top to the bottom, like a waterfall.



WHAT IS AGILE MODEL?

- Software development method.
- Used mainly in advertising agency
- Provides a structured and standardized process for all phases of the system development process.
- Requirements and solutions develop through collaboration between individual organizing, cross functional project teams.
- A conceptual framework that encourages interactions throughout the development

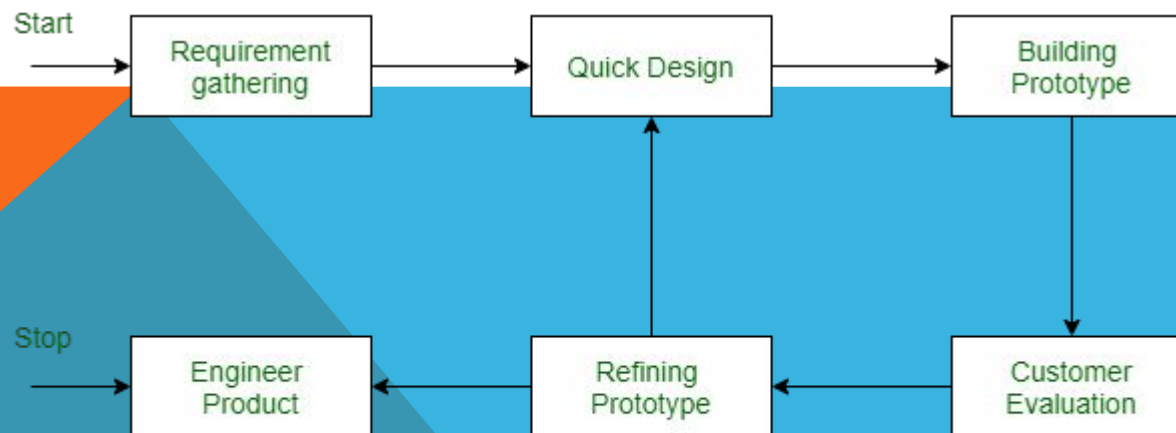


WHAT IS PROTOTYPE MODEL?

- Incomplete versions of the program which is in development.
- Benefits systems that will have many interactions with the users.

Process :

- Identify basic requirements.
- Develop initial prototype , this includes the user interface only.
- Review the customers, include the end-users, examine the prototype and provide feedback on additions or changes.
- Improve the prototype after a process of revision.



PROTOTYPE MODEL

Advantages

- Save time and cost
- Helps to improve and increase user participation

Disadvantages

- Higher risk in miscommunication between user objectives and developer design
- Too much of attachment of the developer
- Too much time spend on development
- Additional cost in rolling out prototype

AGILE MODEL

Advantages

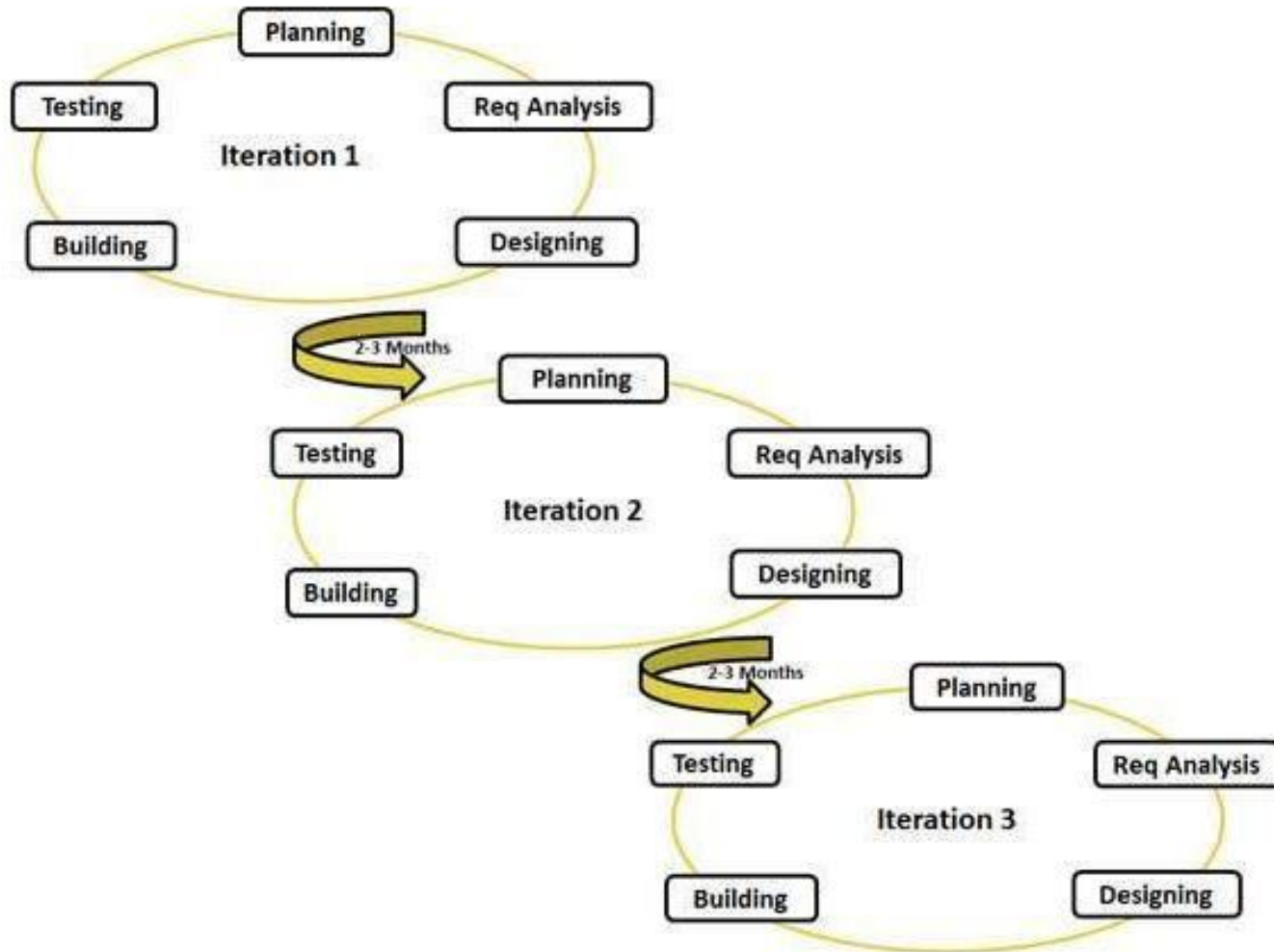
- Encourages interaction, development, teamwork, collaboration and process adaptability
- Minimal planning and phases are in short time frame
- Tests can be conducted during the design cycle
- Less risk and has more flexibilities. Therefore, save time in modifying and developing the information system.

Disadvantages

- Multiple completed phases of development might be required to release a product or new features.
- inefficient in large organizations
- More programmer centric than user centric

PROGRESS FLOW OF AGILE MODEL

Here is a graphical illustration of the Agile Model:



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